

Title of the Paper in English, Be Concise, Specific and Relevant, Bold, Times New Roman, Size 13

Full name (in English Order: EO; for example: Van A Tran NOT Tran Van A) of first author¹ , Full name (in EO) of second author², Full name (in EO) of third author¹

¹Affiliation for first author (just write the name of the university and the country, for example: Ho Chi Minh City University of Technology and Education, Vietnam)

² Affiliation for second author (more detailed information such as department, faculty, or the university's address should not be provided here, but in the authors' biography section)

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KEYWORDS

Keyword 1;

Keyword 2;

Keyword 3;

Keyword 4;

Keyword 5.

ABSTRACT

The abstract must include from 150 to 250 words, NOT include formatted equations. The abstract should be a single paragraph, without abbreviations, comments and references (If references are required, enter the authors name and year in brackets single). It should follow the style of structured abstracts, but without headings and numberings, including the following aspects: 1) Questions and purpose of the study 2) Research methods: Describe the way/method to solve the problem(s), such as develop theory/ methods of data collection, data processing... 3) Results: Summarize the articles main findings, including data that may be taken as essential remarks of the study. The results must be presented in the way that answers the research questions proposed. 4) Conclusion: One or two sentences to conclude the research results and their possible applications. The abstract must provide information briefly but include data to illustrate and get straight to the point. Keywords: immediately after the abstract; provide a minimum of 5 keywords; avoiding general and plural terms and multiple concepts (avoid, for example, 'and', 'of'); be sparing with abbreviations (only abbreviations firmly established in the field may be eligible). These keywords will be used for indexing purposes, separated by a semicolon, end by a dot.

Doi: <https://doi.org/10.54644/jte.2026.xxxx>

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1. Introduction

The **Introduction** provides the background and motivation for the research. It explains *why* the study was conducted, *what problem* it addresses, and *how* it contributes to existing knowledge. A well-written Introduction in IEEE format usually includes **five key elements** as in Table 1.

Table 1. IEEE-Style Introduction Outline.

Part	Content	Example phrase
Background	What is the topic? Why is it important?	“Recently, ... has attracted considerable attention.”
Problem	What is missing or unsolved?	“However, ... remains a significant challenge.”
Related work	What has been done before?	“Several studies have addressed this issue [1]–[3], but ...”

Objective	What does this paper propose or achieve?	“In this paper, we propose ...”
Paper organization	How is the paper structured?	“The rest of this paper is organized as follows...”

This article template let you quickly format your article and help with the placement of specific elements, such as title, author list, abstract, keywords and the paper structure. It would be noted that all manuscripts must contain the essential elements needed to convey your manuscript.

1.1. The article outline

The article must include: Front matter, Maintext/body, Back matter.

1.1.1. Front matter

The front matter has to include: Title, Authors' Fullname and Affiliation, Abstract, Keywords (at least 5)

1.1.2. Maintext/body

The front matter must contain: Introduction, Materials and Methods, Results and Discussions, Conclusions.

The article must be written in the IMRAD structure (Introduction – Methods/Materials – Results – And Discussion). IMRAD structure is a unique, standard structure in international scientific community.

1.1.3. Back matter

The back matter must contain: Acknowledgments, Conflict of Interest, Data Availability Statement, References, Authors' Biography.

1.2. The article layout

1.2.1. The article format

The layout format must be set in A4 size, portrait layout, with specific Page Setup parameters as follows: Top: 3.0 cm, Bottom: 2.5 cm, Left: 3.0 cm, Right: 2.0 cm, Header from Top: 1.0 cm, Footer from Bottom: 1.2 cm. Article content uses 11-point Times New Roman font.; line spacing: single, before 0; after 3; justified margin. Figure 1 shows the demonstration of JTE article layout.

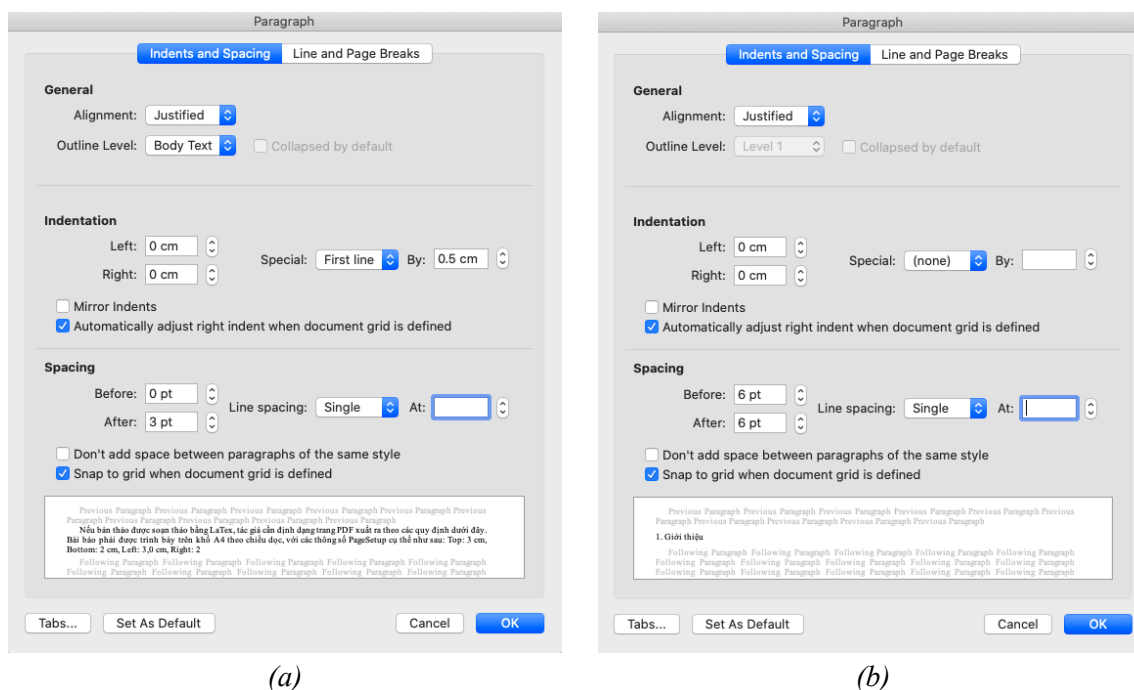


Figure 2. Paragraph setup for (a): Content text and (b) Heading of any section

1.2.3. The article length

The paper must not exceed 10 pages in length (exclude the references and authors biography). If the content exceeds this limit, less essential materials should be placed in a separate supplementary file. The supplementary information should be made available online through a link provided in the *Data Availability Statement* section.

1.2.4. The article title

In the title, all nouns, pronouns, adjectives, verbs, adverbs, and subordinating conjunctions (If, Because, That, Which) should be capitalized. Capitalize abbreviations that are otherwise lowercase (i.e., use DC, not dc or Dc) except for unit abbreviations and acronyms. Articles (a, an, the), coordinating conjunctions (and, but, for, or, nor), and most short prepositions are lowercase unless they are the first or last word. Prepositions of more than three letters (Before, From, Through, With, Versus, Among, Under, Between, Without) are capitalized.

Examples:

- Nonlinear Gain Coefficients in Semiconductor Lasers: Effects of Carrier Heating
- Self-Pulsation in an InGaN Laser—Part I: Theory and Experiment

1.2.5. The article structure

The article must be written in the IMRAD structure (Introduction – Methods/Materials – Results – And Discussion). IMRAD structure is a unique, standard structure in international scientific community.

1.3. References, figure, and table

References should be numbered in the order of their appearance in the text and indicated by numerals in square brackets, e.g., [1], [2], [3], The list of references should be arranged in the same numerical order as they are cited in the text. All figures and tables must also be referred to in the text in numerical order.

1.3.1. References

References are **numbered in the order of their first appearance** in the text. Each reference number is enclosed in **square brackets**, e.g., [1], [2], [3]. Multiple references are grouped together: [1]–[3] or [1], [4], [5]. The title of the section is “REFERENCES” in all caps and centered. Journal and book titles are in *italics*, article titles in regular font.

For example:

- [1] J. M. Smith and K. Tan, “A survey of wireless sensor networks,” *IEEE Communications Surveys & Tutorials*, vol. 15, no. 3, pp. 987–1005, Sep. 2021.
- [2] A. Brown, *Modern Control Systems*, 3rd ed., New York, NY, USA: McGraw-Hill, 2019.

1.3.2. Figures

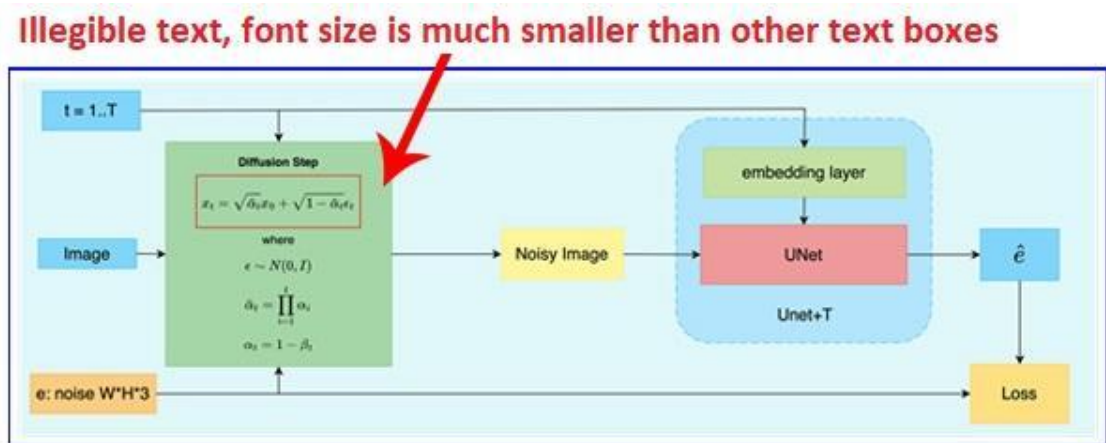
Figures should be **numbered consecutively** in the order they are mentioned in the text: Figure 1, Figure 2, Figure 3, etc. Each figure must be **cited in the text** before or immediately after it appears. Figure captions are placed **below the figure**, centered, in **10-point Times New Roman** with line spacing: single, before 6; after 6

Example:

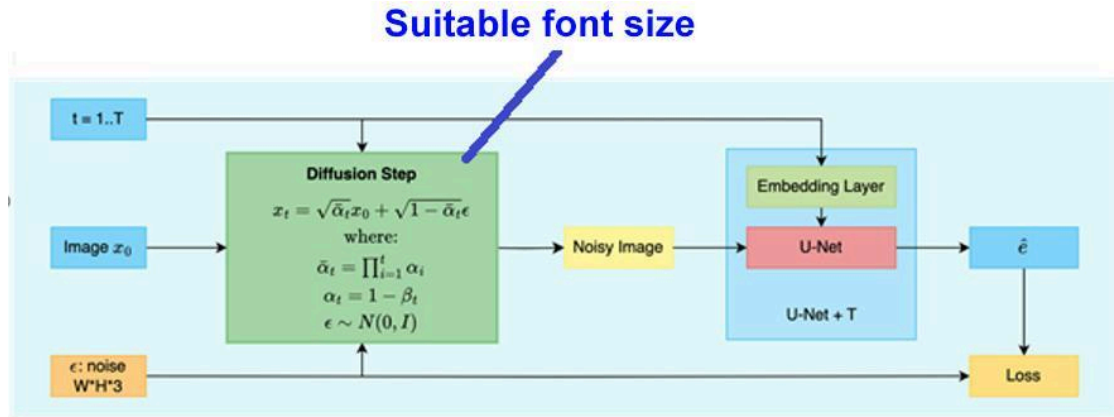
Figure 3. *Illustrative images based on ethical standard A.2.*

Figures should be **clear, legible, and of high resolution (≥ 300 dpi)**, use **black and white** unless color is necessary. All text in figures (labels, legends, etc.) should be readable at journal size.

According to ethical standard A.2, the information in the article must be presented concisely, clearly, and succinctly. Figures in the article should not contain areas without information or with unnecessary details. Any text (if present) within the figures must have a readable font size that is proportional to the font size of the article (unless the author intends to emphasize a specific point). Composite images (for comparison purposes) should have consistent reference frames. The examples below illustrate figures that meet and do not meet these criteria. An example of acceptable and unacceptable figures is shown in Figure 3.

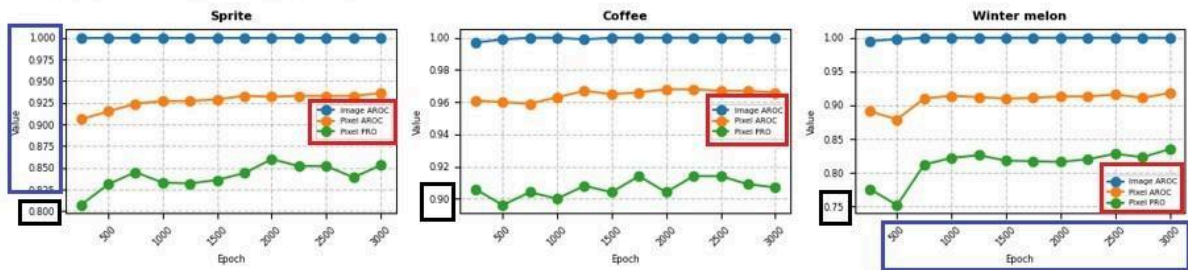


(a) Font size is not proportional.



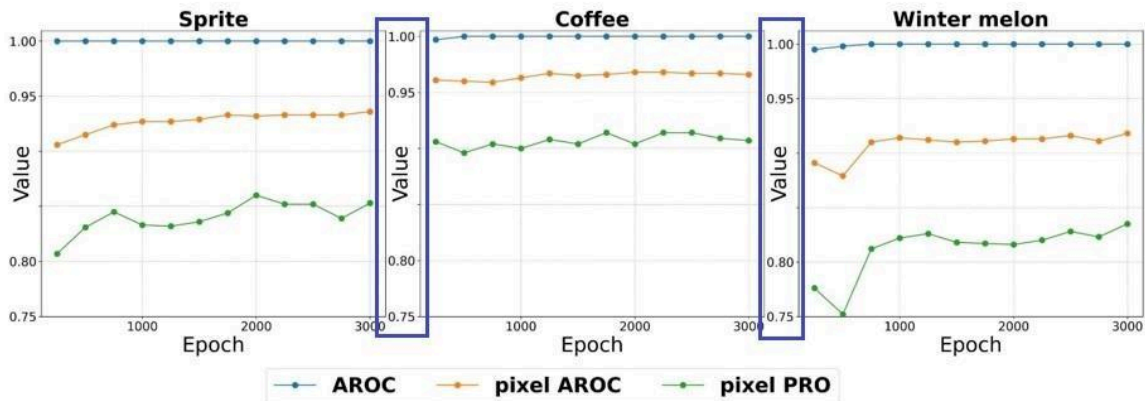
(b) Well-balanced font size.

Mismatched y-axis range of values, small text of legend and ticks, hard to read



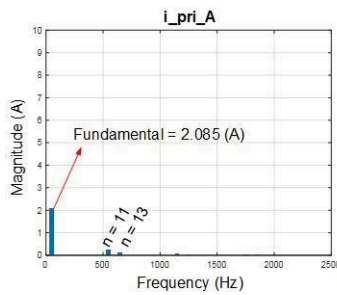
(c) Y-axis values are mismatched.

The figure basically meets the requirements, but it could be more concise by using a shared Y-axis for the three subplots (merging them into a single horizontal layout)

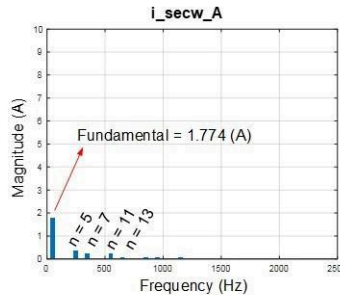


(d) Consistent Y-axis values make it easier for readers to compare and contrast the data.

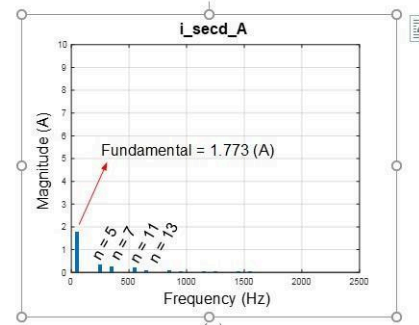
The data falls within a narrow range: X-axis (1–1500), Y-axis (0–2), but the plot frame is large: X-axis (1–2500), Y-axis (0–10)
→ this leads to wasted page space.



(a)



(b)



(c)

(e) The frame size is disproportionate to the content.

only sv The borders of the figure should fit closely around the content, avoiding large blank spaces when
S₁, S₂ & Crop your figure along solid lines, not dash lines

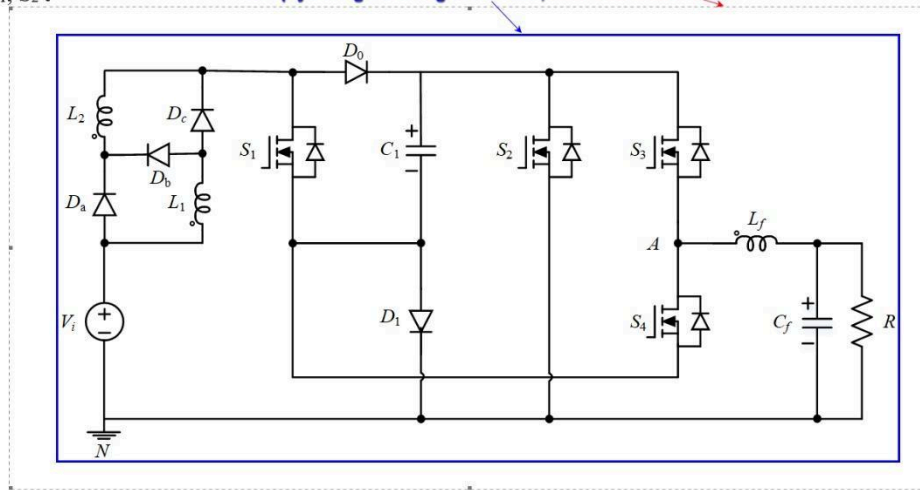
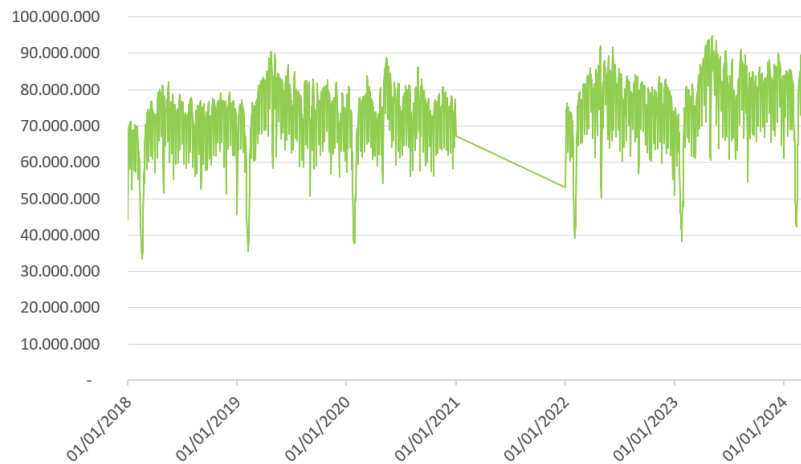
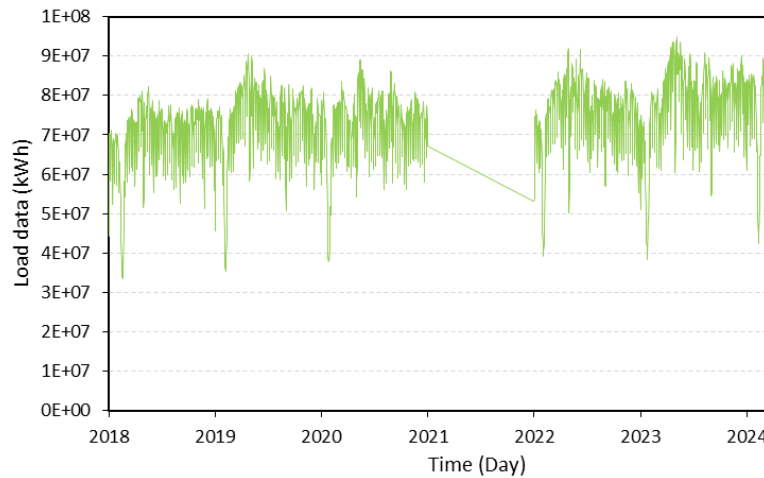


Figure 1. Single-phase four-switch current-fed inverter topology

(f) The image is not tightly cropped to the content area, resulting in excessive empty space (above Figure 1), which makes editing and layout more difficult.



(g) y-axis value is long and not scientific shown.



(h) y-axis and x-axis are suitable shown.

Figure 3. Illustrative images based on ethical standard A.2

(a), (c), (e), (f), and (g): unacceptable images.

(b), (d), and (h): acceptable images.

1.3.3. Tables

Tables are **numbered consecutively** and cited in the text in numerical order: Table 1, Table 2, Table 3, etc. Table titles are placed **above the table**, left-aligned, in **10-point Times New Roman**, with line spacing: single, before 6; after 6

Example:

Table 2. IEEE-Style Abbreviation and Acronym Guidelines.

The title should be **concise and descriptive**. Every **figures or table must be mentioned in the text** in ascending order. Avoid placing figures or tables before their first mention.

Note: components of a table must be in editable text, not in the form of screenshots. Table Properties have Table tab with Preferred Width = 16 cm, Table options with Top = 0 Bottom = 0 Left = 0,05 Right = 0,05 cm, Row tab with Not check Specify height. Font size of text inside table = 10, Layout of Table with Spacing Before = 3pt After = 3pt, Line Spacing: Single. No borders for the left and right edges of the table; the first row of the table (which is the title of each column) should not be left blank or missing information.

1.4. Equations

Important equations should be numbered consecutively throughout the paper. Only equations that are referenced in the text need to be numbered. When referring to an equation, use only the number in parentheses:

- Correct: *as shown in (1)*
- Correct (at the beginning of a sentence): *Equation (1) shows that ...*
- Incorrect: *as shown in Eq. (1)* (avoid “Eq.” in the middle of a sentence)

The equation number should be **right-aligned** and enclosed in parentheses, line spacing: single, before 6; after 6. For example:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

An example of figures, tables, section heading and subheading, equation is shown in the Figure 4.

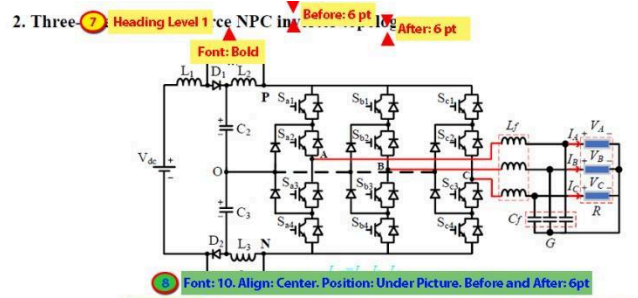


Figure 1. Three-level quasi-Z-source NPC inverter

2.1. Three-level quasi-Z-source NPC inverter

As illustrated in Fig. 1, the topology consists of the MqSB network which is formed by two identical impedance networks including two inductors (L_1, L_2), two capacitors (C_1, C_2), four diodes (D_1, D_2, D_3, D_4), two active switches (T_1, T_2) and the traditional NPC structure.

2.1.1. Zero state

load voltage in this state is zero. Therefore, the ST state is generated in the zero vector in order not to cause the distortion at output voltage.

Table 1. Switching states of the proposed inverter

Mode	Switches	Diodes	Output Voltage
NST	S_{X1}, S_{X2}	D_1, D_2, D_3, D_4	$+V_{PN}/2$
	S_{X2}, S_{X3}		0
	S_{X3}, S_{X4}		$-V_{PN}/2$
ST	$S_{X1}, S_{X2}, S_{X3}, S_{X4}$	D_1, D_4	0

2.1.1.1. NST mode

Considering the non-short circuit state as shown in Figure 2.7(a), the diodes D_1 and D_2 are forward biased, the two capacitors C_2 and C_3 are stored energy from the input power V_{dc} and the voltage of the inductor L_1 . Meanwhile, capacitors C_1 and C_4 store energy from inductor L_2 and L_3 . The duration of this state is $(1-D_{ST})T$. The voltage across inductors L_1, L_2 , and L_3 are obtained as following equations:

$$\begin{cases} V_{PN} = V_{L1} + V_{C2} + V_{C3} \\ V_{L2} = -V_{C1} \\ V_{L3} = -V_{C4} \end{cases} \quad (1)$$

2.1.1.2. ST mode

Considering the ST mode as shown in Figure 2.7(b) the switches of the inverter (S_{1x}, S_{2x}, S_{3x} , and S_{4x})

Figure 4. Guidelines of figures, of tables, of section heading and subheading.

1.5. Abbreviation and acronym

When an abbreviation or acronym first appears in the text, **spell out the full term**, followed by the abbreviation in parentheses. Thereafter, you can use the abbreviation alone.

For example: Orthogonal Frequency Division Multiplexing (OFDM) is widely used in modern communication systems. OFDM provides high spectral efficiency and robustness against multipath fading.

Regulations of abbreviation or acronym are summarized as Table 2.

Table 2. IEEE-Style Abbreviation and Acronym Guidelines.

Rule	Description	Example
Define on first use	Full term + (abbreviation)	“Support Vector Machine (SVM)”
No need to define common terms	Use as is	“LED”, “GHz”, “IEEE”
Keep consistent	Same acronym throughout	Always use “SVM”, not “Support VM”
Plurals	Add “s” only	“CPUs”, not “CPU’s”
Titles & abstracts	Avoid or define immediately	“... using an ANN-based approach”

1.6. Unit and number

It must be consistently used International System of Units (SI Units) for the data in the article. Other kinds of unit should be converted to SI units whenever possible. Italic format is applied for symbols of calculated quantities. Decimals presented in Vietnamese articles by "," and by "." in English articles. Use a **space** between the number and the unit symbol. For example:

✓ Correct: 10 mA, 5 V, 20 °C

✗ Incorrect: 10mA, 5V, 20°C

2. Materials and Methods

This part should be described in detail the ways or methods, materials, devices necessary to solve the research problems. They should be described with sufficient detail so that other researchers can replicate and build on published results. New methods and protocols should be described in detail while well-established methods can be briefly described and appropriately cited. The approach might be by theoretical developing, empirical researching, surveying, etc. Applied approach should be explained the advantages and, if possible, be compared with previous studies. In the case of theoretical developing research, it should be present theoretical bases to find out solution to research problems.

3. Results and Discussion

This section shows the results obtained from doing research. Images or tables could be used to illustrate the results and support for discussions. Authors should discuss the results and how they can be interpreted in perspective of previous studies and of the working hypotheses. The findings and their implications should be discussed in the broadest context possible and limitations of the work highlighted. Future research directions may also be mentioned.

If necessary, one part of the article might be divided into several sections. It is possible to provide information introducing small parts between the main heading of the section and the subheading. Subheadings (Heading level 2) are in size 11, Bold and Italic as following.

4. Conclusions

This section summarizes important outcomes drawn from the section 3: Results and Discussion. Note to avoid duplicating Abstract. It could be orientations of researching, developing and applying research results.

Acknowledgments

All of contributing individuals and organizations of the study should be disclosed. The sponsors of the research are shown here. For example, authors can write: "This work belongs to the project in 2022 funded by the Ho Chi Minh City University of Technology and Education, Vietnam"

Conflict of Interest

Authors must confirm "no conflict of interest", write down as "The authors declare no conflict of interest."

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to English language editing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability Statement

If the article exceeds ten pages, the excess part will be written as supporting information. You can display your supporting information on your own or institutional website. The authors describe the way to provide the readers supporting information here. Authors can write: "The data that support the findings of this study are available from the corresponding author upon reasonable request."

REFERENCES

Please ensure that every reference cited in the text is also present in the reference list (and vice versa). Any references cited in the abstract must be given in full. References must be numbered in order of appearance in the text and listed individually at the end of the manuscript. We recommend preparing the references with a bibliography software package, such as EndNote, Mendeley, or Zotero... or use "Insert Citation" in MS Word to avoid typing mistakes and duplicated references.

Citations in the text should follow the referencing style of IEEE (see detail the reference guide at <https://ieeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>). Some important notes are given below.

References need not be cited in the text. When they are, they appear on the line, in square brackets, inside the punctuation. Grammatically, they may be treated as if they were nouns or footnote numbers, e.g.: According to [1]; as demonstrated in [2]; as shown by Brown [4], [5]; as mentioned earlier [2], [4], [5], [6], [7], [9]; Smith [4] and Brown and Jones [5]; Wood et al. [7]. Reference ranges in text will not include an en dash. All references will be written out. For example, "[1]–[4]" will now be "[1], [2], [3], [4]".

****NOTE:** Use "et al." when three or more names are given for a reference cited in the text

References must be **numbered in order of appearance in the article**. The order of references should be manual typed, not used numbered list of Word, have Paragraph properties with Indentation Left = 0, Special = Hanging By = 0,75 cm. References have font size 8, Justify Alignment, Spacing Before and After = 0

All references must be presented in Romantic characters, i.e. in English.

Some common examples of references in IEEE format are demonstrated as below: [1] for article without DOI ([2] is equivalent example), [3] for article with DOI ([4] is equivalent example), [5] for article with ID ([6] is equivalent example), [7] for article in other Language ([8] is equivalent example), [9] for article with url (for example), [10] for unpublished paper (for example), [11] for article in conference proceedings without DOI ([12] is equivalent example), [13] for article in conference proceedings with DOI ([14] is equivalent example), [15] for Paper Presented at a Conference ([16] is equivalent example), [17] for book, [18] for book with chapter title ([19] is equivalent example), [20] for patent, [21] for dissertations (for example), [22] for standards (for example):

- [1] J. K. Author, "Name of paper," *Abbrev. Title of Periodical*, vol. x, no. x, pp. xxx-xxx, Abbrev. Month, year.
- [2] T. T. C. Duong, C. C. Nguyen, and T. D. Tran, "Synchronization Sliding Mode Control of Closed-Kinematic Chain Robot Manipulators with Time-Delay Estimation," *Appl. Sci.*, vol. 12, no. 11, p. 5527, 2022.
- [3] J. K. Author "Name of paper," *Abbrev. Title of Periodical*, vol. x, no. x, pp. xxx-xxx, Abbrev. Month, year, doi: xxx
- [4] G. R. Sheu *et al.*, "Distribution of atmospheric mercury in northern Southeast Asia and South China Sea during Dongsha Experiment," *Atmos. Environ.*, vol. 78, pp. 174–183, Oct. 2013, doi: 10.1016/J.ATMOSENV.2012.07.002.
- [5] J. K. Author, "Name of paper," *Abbrev. Title of Periodical*, vol. x, no. x, Abbrev. month, year, Art. no. xxx.
- [6] J. Zhang and N. Tansu, "Optical gain and laser characteristics of InGaN quantum wells on ternary InGaN substrates," *IEEE Photon. J.*, vol. 5, no. 2, Apr. 2013, Art no. 2600111.
- [7] J. K. Author, "Name of paper," (in Language), *Abbrev. Title of Periodical*, vol. x, no. x, Abbrev. month, year, Art. no. xxx
- [8] E. P. Wigner, "On a modification of the Rayleigh–Schrodinger perturbation theory," (in German), *Math. Naturwiss. Anz. Ungar. Akad. Wiss.*, vol. 53, p. 475, 1935.
- [9] J. K. Author, "Name of paper," *Abbrev. Title of Periodical*, vol. x, no. x, pp. xxx-xxx, Abbrev. Month, year. Accessed: Month, Day, Year, doi: 10.1109.XXX.123456. [Online]. Available: site/path/file
(for example: [9] W. P. Risk, G. S. Kino, and H. J. Shaw, "Fiber-optic frequency shifter using a surface acoustic wave incident at an oblique angle," *Opt. Lett.*, vol. 11, no. 2, pp. 115–117, Feb. 1986. [Online]. Available: <https://opg.optica.org/ol/abstract.cfm?uri=ol-11-2-115>)
- [10] J. K. Author, "Title of paper," unpublished.
(for example: [10] B. Smith, "An approach to graphs of linear forms," unpublished.)
- [11] J. K. Author, "Title of paper," in *Abbreviated Name of Conf.*, (location of conference is optional), (Month and day(s) if provided) year, pp. xxx-xxx.
- [12] T. V. Do, T. L. P. Nguyen, T. T. T. Nguyen, L. T. H. Lam, P. S. Thanh, and T. D. Nguyen, "Ankle joint rehabilitation system-the preliminary results," in *Proceedings of the Sixth International Conference on Research in Intelligent and Computing*, 2021, pp. 33–37.
- [13] J. K. Author, "Title of paper," in *Abbreviated Name of Conf.*, (location of conference is optional), year, pp. xxx-xxx, doi: xxx.
- [14] G. Veruggio, "The EURON roboethics roadmap," in *Proc. Humanoids '06: 6th IEEE-RAS Int. Conf. Humanoid Robots*, 2006, pp. 612–617, doi: 10.1109/ICHR.2006.321337.

- [15] J. K. Author, "Title of paper," presented at the Abbreviated Name of Conf., City of Conf., Abbrev. State, Country, Month and day(s), year, Paper number.
- [16] K. C. Dinh, N. S. Dao, H. D. Le, H. L. Le, T. D. T. Cam, and D. T. Tran, "Kinematics and Dynamics for a 4-DOF Parallel Robot," presented at the 2021 International Conference on System Science and Engineering (ICSSE), 2021.
- [17] B. Klaus and P. Horn, *Robot Vision*. Cambridge, MA, USA: MIT Press, 1986.
- [18] J. K. Author, "Title of chapter in the book," in *Title of Published Book*, X. Editor, Ed., City of Publisher, State (only U.S.), Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx-xxx
- [19] T. Ogura, "Electronic government and surveillance-oriented society," in *Theorizing Surveillance: The Panopticon and Beyond*, Culloington, U.K.: Willan, 2006, ch. 13, pp. 270-295.
- [20] J. K. Author, "Title of patent," Country Patent xxx, Abbrev. Month, day, year.
- [21] J. K. Author, "Title of dissertation," Ph.D. dissertation, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.
(for example: [21] J. O. Williams, "Narrow-band analyzer", Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, USA, 1993.)
- [22] *Title of Standard*, Standard number, date.
(for example: [22] *IEEE Criteria for Class IE Electric Systems*, IEEE Standard 308, 1969.)
- [23] A. Harrison, *private communication*, May 1995.

First Author Full Name (the same as authors' names section). The first paragraph may contain author's educational background is listed. The degrees should be listed with type of degree in what field, which institution, city, state, and country, and year the degree was earned. The second paragraph lists work experience (including being awarded prizes, honors,...), the previous job and the current job must have year and location (Lab, Department, Faculty, University, Country).

The third paragraph lists all research interests as detailed as possible (keywords).

The last paragraph is author's contact (at least, including email address) Email: abcd@hcmute.edu.vn.

ORCID:  <https://orcid.org/0000-000X-XXXX-XXXX>. Tel (of CoAuthor): 090XXXXXXX

Second Author: Authors' Biography was included in a table with one columns as this demonstration. The Table Properties with Preferred Width = 16 cm, Table Option Top = 0, Bottom = 0, Left = 0,05 cm Right = 0,05 cm, No Border. All authors must verify their authorship of the article using their ORCID, so the ORCID of each author must be provided here.

Email: abcd@hcmute.edu.vn. ORCID:  <https://orcid.org/0000-000X-XXXX-XXXX>

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